

NetSage AI

Human-in-the-Loop Network Troubleshooting Assistant

Submission-ready project report | Verified 30 August 2026 UTC

Abstract

NetSage AI is a demonstrator for Cisco-style Packet Tracer/lab network troubleshooting. It combines deterministic checks with an AI-style structured diagnosis and mandatory human review. The repository contains 32 cases. Final verification completed successfully; 2 unit tests passed. AI-style diagnosis agreed with expected faults in 12/32 cases (37.5%), so this is a human-supervised prototype, not a production-accuracy model.

Architecture

cases.csv and human_review_log.csv feed the rule checker and diagnosis layer. The workflow emits structured diagnoses, compares expected versus predicted faults, joins review decisions, and writes ai_diagnoses.csv, case_results.csv, and dashboard/summary.csv.

Rule checks

duplicate IP, wrong subnet mask, gateway mismatch, interface down, missing VLAN, and missing route.

Evaluation

Review status: Accepted 21, Edited 8, Rejected 3. Agreement: yes 12, no 20. Severity: high 16, medium 12, low 4. Issue types: Routing 6, ACL 4, DHCP 4, DNS 4, Gateway 4, Wireless 4, NAT 3, VLAN 3.

Limitations and future work

The AI-style layer is keyword-driven and not a trained model. The 37.5% agreement rate requires human oversight. Future work includes expanded tests, broader labeled evaluation, improved taxonomy matching, confidence calibration, and a browser dashboard.

How to run

```
cd /home/user/NetSage-AI
python src/rule_checker.py --cases data/cases.csv --limit 3
python src/workflow.py --cases data/cases.csv --reviews data/human_review_log.csv --out outputs
--dashboard dashboard
python -m unittest discover -s tests -p 'test_*.py'
```